ines

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Chapter 1

Design decisions of the ines spec

1.1 Sets

A *set* groups certain entities and gives them common parameters for common constraints.

1.2 Temporal structure

A lot of the preprocessing of the temporal structure is defined by the *solve_pattern*.

The *solve_mode* dictates whether the model uses a single time horizon or a rolling time horizon (single by default). Rolling time horizon means that the same model is used for different horizons and that the input of the next horizon depends on the results of the current horizon. This is not reflected in the model formulation but rather in an algorithm.

The time horizon of every solve pattern is described by a *start_time* and a *duration*. It is sufficient to only give a *duration* value as it starts from the start of the horizon(?) If the solve mode is a rolling horizon, in addition you define the *rolling_horizon* which is also a duration but is smaller than the *duration* of the solve pattern. Each roll will have a length equal to the rolling horizon and will roll as many times as needed to reach the duration of the solve pattern. You can also define a *rolling_jump* to have a gap between each roll. The ines spec currently does not allow for variable roll durations and jumps as the only accepted value is a duration value.

includes_solve_pattern allows for nested setups. For example, you can combine a rolling model hydro model with a rolling dispatch model. When the hydro model includes the dispatch model it means that first the hydro model will roll, then the dispatch model will roll as many times within the rolling horizon of the hydro model (assuming the rolling horizon of the dispatch model is smaller than the rolling horizon of the hydro model).

For rolling and nested patterns to communicate with each other, there are some parameters in the nodes, connections and units to set the method to bind them by various solves (i.e. Benders decomposition). However, the parameters for which this holds is not complete (e.g. a method value 'fix based on upper level').

The *time_resolution* indicates how the data should be sampled. For example, hourly data can be sampled as two hourly data. Different resolutions in different parts of the system are realised with a *set*. The *time_resolution_scope* indicates whether the data has different time resolutions defined by sets or not. The time resolution of the solve pattern then needs to have the smallest possible resolution to be able to cover all time steps in the *timeline* (see later).

period is meant for a larger time scale, e.g. an investment time scale. It is defined by a *start_time* and a *years_represented*. Typically expressed in years, but sometimes there are also quarterly investments that is less than a year. $p_{period}^{years\_represented}$ is used to derive the time steps that belong to a period. The used periods also need to be defined in the solve pattern.

The *system* entity has a parameter called the *timeline* and holds all of the allowed datetimes in the model. A lot of the conversion tools expect this timeline to be there. The timeline is a timeseries with each datetime representing a timestep and each value representing the duration which is a float that is to be interpreted as hours (even though one could expect to use a map with datetimes and durations, however, most tools use hours and it would make the conversion tools more difficult to develop).

1.3 Stochastic structure

The stochastic structure is organised as an alternative parameter value, i.e. you can either provide a deterministic value (e.g. $profile\_limit\_lower$) or a forecast value (e.g. $profile\_limit\_lower\_forecast$). The forecasts can either be independent or branching. The branching forecasts require an additional dimension (i.e. a distinction between analysis time and forecast time?). The *stochastic_scope* method parameter (in the solve pattern) determines which stochastic structure the model uses. *whole_model* implies that the entire model uses the same stochastic structure and every parameter needs to be stochastic if it can be. *single_set* implies that the entire model uses the same stochastic structure but not all parameters need to be stochastic. *multiple_sets* implies that different parts of the model use different stochastic structures.

1.4 Economic representation

The ines spec assumes that the distribution of annuities is flat, i.e. there is no lead time. M_{period} is the milestoneyear of the period; could be start of the simulation for all periods.

$$annuity_{unit,period} = \frac{p_{unit,period}^{interest\_rate}}{1-(1+p_{unit,period}^{interest\_rate})^{-p_{unit}^{lifetime}}}$$

$$discount\,factor_{unit,period} = \sum_{y=Y_{period}}^{Y_{period}+p_{unit}^{lifetime}} \frac{1}{1-(1+p_{unit,period}^{interest\_rate})^{y-M_{period}}}$$

$$salvage\,factor_{unit,period} = \begin{cases} 0 & Y_{end} \geq Y_{period} + p^{lifetime}_{unit} \\ \sum_{y=Y_{end}}^{Y_{period}+p^{lifetime}_{unit}} \frac{1}{(1+p^{interest\_rate}_{unit,period})^{y-M_{period}}} & Y_{end} < Y_{period} + p^{lifetime}_{unit} \end{cases}$$

$$p^{economic\_investment}_{unit,period} = annuity_{unit,period} \cdot (discount\,factor_{unit,period} - salvage\,factor_{unit,period})$$

$$p^{economic\_fixed}_{unit,period} = \begin{cases} \frac{p^{years\_represented}_{period}}{1-(1+p^{interest\_rate}_{unit,period})^{Y_{period}-M_{period}}} & p^{years\_represented}_{period} < 1 \\ \sum_{y=Y_{period}}^{Y_{period}+p^{years\_represented}_{period}} \frac{1}{1-(1+p^{interest\_rate}_{unit,period})^{y-M_{period}}} & p^{years\_represented}_{period} \geq 1 \end{cases}$$

$$p^{economic\_other}_{unit,period} = \begin{cases} \frac{1}{1-(1+p^{interest\_rate}_{unit,period})^{Y_{period}-M_{period}}} & p^{years\_represented}_{period} < 1 \\ \sum_{y=Y_{period}}^{Y_{period}+p^{years\_represented}_{period}} \frac{1}{1-(1+p^{interest\_rate}_{unit,period})^{y-M_{period}}} & p^{years\_represented}_{period} \geq 1 \end{cases}$$

1.5 units, nodes, links and flows

Units and links operate between nodes. A node has a *node_type* which indicates how the node is balanced. *balance* implies that all incoming and outgoing flows need to equal each other. *commodity* implies the use of a commodity to deal with the imbalance of flows in the node. A typical configuration is a 'source' node with a *commodity* node type connected to a unit (through a *node__to_unit* entity) and the unit connected to a 'demand' node with a *balanced* node type (through a *unit__to_node* entity). The 'demand' node would then also have a parameter *demand* which denotes the demand of the node in every time step.

The efficiency etc. is defined at the unit level, whereas the capacity is defined at the flow level (although the flow can also define an efficiency that is more flexible, similar to SpineOpt). The capacity needs to be defined for every flow. The efficiency needs to be defined for every unit. The capacity is per unit as it is multiplied with the number of existing units, which means that both the capacity and the number of existing units always need to be specified. The direction of the flow is handled by *node__link__node*, on one hand in the transfer method in *link* and on the other hand with multiplier from *constraint_flow*. To deal with multiple profiles for a unit in a multiyear setup, you can define a unit with the same parameters but different profiles (though in some setups you would require a map with a period and a time, which is currently not supported).

unit_conversion_method provides a few conversion methods including *partial_load_efficiency* and *piecewise_linear* as depicted in the figure below.

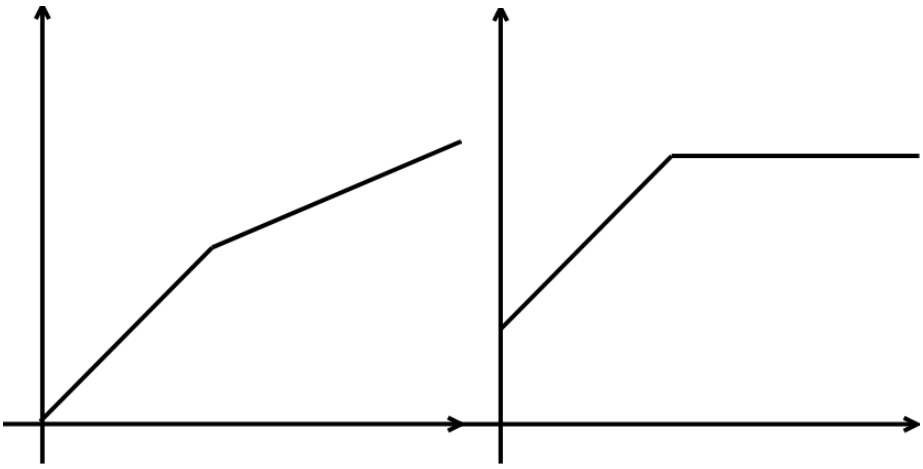

Figure 1.1: Visual depiction of *unit_conversion_method*: piece wise linear (left) and partial load efficiency (right)

To define a cyclic condition for storage in a node, you set the *storage_binding_method* to leap over a period or solve. You can also fix the initial state of the storage with the *storage_fix_state_method*.

Across the entities there are some scaling parameters (methods) that can be used to, e.g., scale the flow of a partial solve to a year.

Some entities also have methods for investments. When it is cumulative, it describes over multiple periods what is the total allowed capacity. Which can also be set for a set.

To set up a user constraint you define a *constraint* with the *sense* and *constant*. You can then link a *unit___to_node* entity to this constraint through the *constraint_flow_coefficient*.

Chapter 2

Proof of concept for the certification process

As the ines spec describes an interoperable data format, it is particularly interesting to define a certification process in that format such that it can be used to certify different energy system models for the correct implementation of certain features such as, e.g., different time resolutions in different parts of the model.

Here, we only consider a proof of concept to show the potential of certification with the ines spec. Once a consortium has been formed around the ines spec, it makes sense to continue the development of the certification process. In the following, we cherry pick some setups that show different aspects of energy system models. Some suggestions for expansions on these setups can be found in appendix C.

While it is possible to describe different features in the ines spec, there is not actually a specific model associated to it. Therefore, if we want to obtain the exact solution for the certification process, we'll have to solve the system manually. In the following, not only do we provide a description of the setup but also provide the specific equations to solve that setup.

Figure 2.1 shows the workflow in Spine Toolbox and Table 2.1 gives a brief overview of the setups.

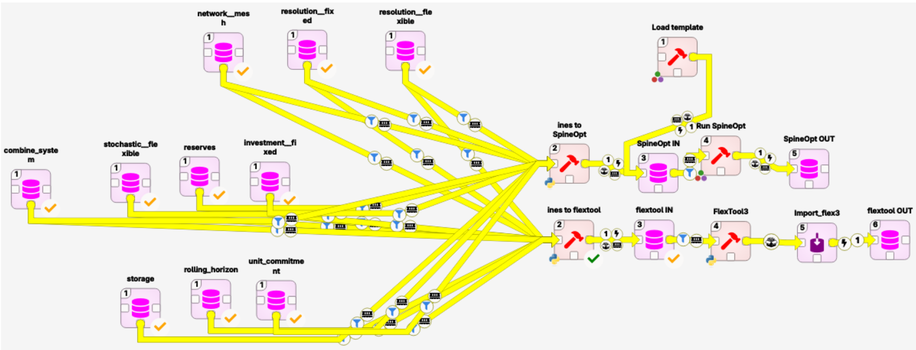

Figure 2.1: Certification workflow

name	description
network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular (but one directional) network with the two supply units and demand unit. The cheapest unit should be chosen.
resolution = fixed	network = meshed but expanded to a time series. Both units have the same resolution.
resolution = flexible	resolution = fixed but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
stochastic = flexible	resolution = fixed but there are scenarios for the demand and the resource prices.
reserves	resolution = fixed but there is a demand for reserves. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
unit commitment	resolution = fixed but there is unit commitment for both units. There is a minimum up time, a minimum down time and a part load efficiency.
investment = fixed	resolution = fixed but there are investment variables.
storage	resolution = fixed but with storage. Storage follows a delta formulation with intra and inter time steps and blended representative periods.
rolling horizon	investment = fixed but there are two horizons and the model rolls from one horizon to the other. The demand is increased in the second horizon.
combined system	A combination of all the previous setups. Sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. There are stochastics for the renewables and import prices.

Table 2.1: Expansion of the certification process.

2.1 network = mesh

The goal of the setup is to test the energy balance in space. To that end the demand and supply are scattered over different nodes. They are connected to each other through directional lines with small losses but sufficient capacity. The lines are directional to avoid circular flows and the lines have losses to avoid multiple possible solutions. One of the supply units is obviously cheaper such that it is very clear what the solution should be.

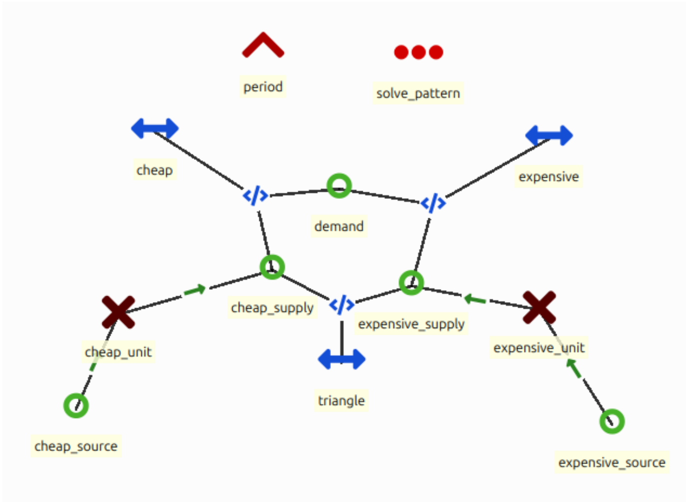

Figure 2.2: Implementation of network = mesh in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = p_{cheap\_source}^{other\_operational\_costs} \cdot v_{cheap\_source}^{flow} + p_{expensive\_source}^{other\_operational\_costs} \cdot v_{expensive\_source}^{flow}$$

subject to:

$$\begin{aligned}
p_{demand}^{flow_profile} &= v_{cheap_link,out}^{flow} + v_{expensive_link,out}^{flow} \\
v_{cheap_link,out}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in}^{flow} \\
v_{expensive_link,out}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in}^{flow} \\
v_{triangle,out}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in}^{flow} \\
v_{cheap_link,in}^{flow} &= v_{cheap_supply}^{flow} + v_{triangle,out}^{flow} \\
v_{expensive_supply}^{flow} &= v_{expensive_link,in}^{flow} + v_{triangle,in}^{flow} \\
v_{expensive_supply}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source}^{flow} \\
v_{cheap_supply}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source}^{flow}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in} \\
0 \leq & v^{flow}_{cheap\_link,out} & \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in} \\
0 \leq & v^{flow}_{expensive\_link,out} & \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in} \\
0 \leq & v^{flow}_{triangle,out} & \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply} & \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source} \\
0 \leq & v^{flow}_{cheap\_supply} & \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source}
\end{aligned}
$$

2.2 resolution = fixed

The goal of the setup is to test the energy balance in time. To that end, the setup starts from network = meshed and adds a time series for the demand and the supply. All entities have the same resolution.

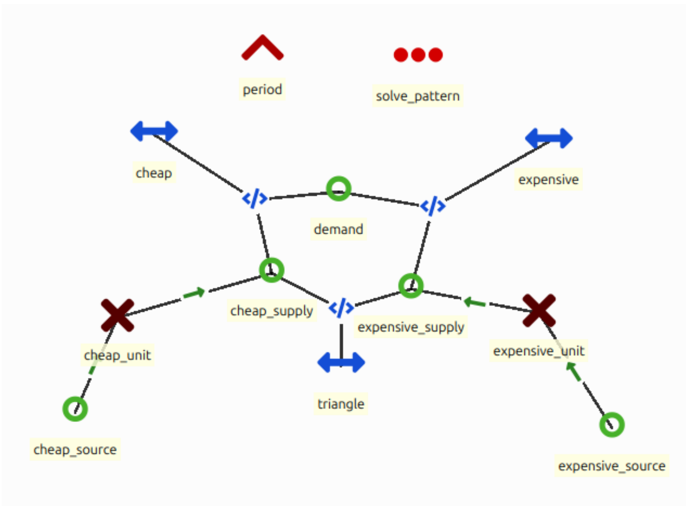

Figure 2.3: Implementation of resolution = fixed in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
z = \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source,t} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

2.3 resolution = flexible

The goal of this setup is to test the energy balance in time and space. To that end, the setup starts from resolution = fixed but reduces the resolution of one of the supply units. The resolution is only changed at the source such that the supply of the unit is still able to provide the same output as before.

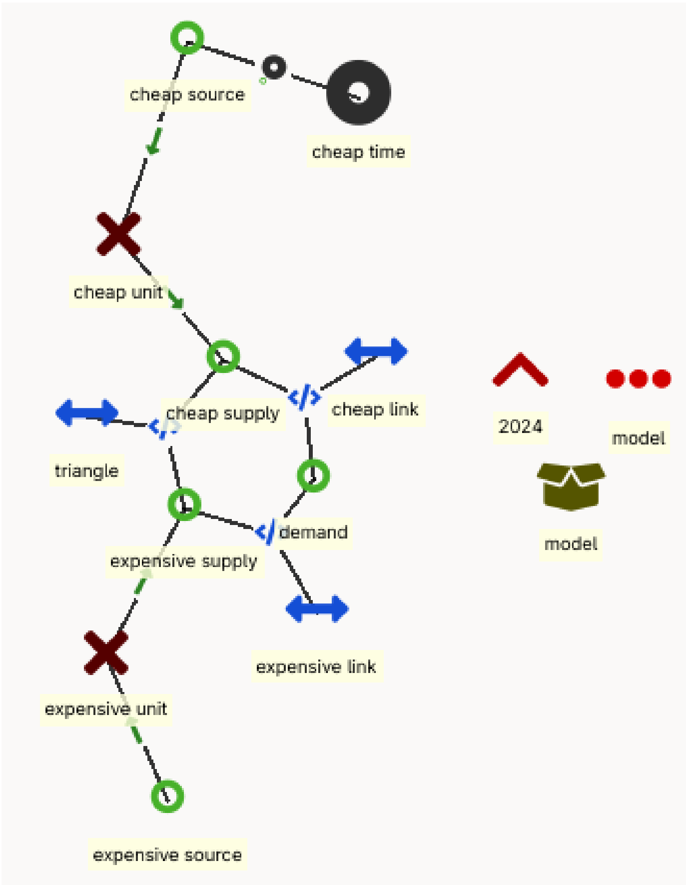

Figure 2.4: Implementation of resolution = flexible in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_{t \in t_{cheap}} p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + \sum_{t} p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} &= v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
v^{flow}_{cheap\_link,out,t} &= p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} &= p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
v^{flow}_{cheap\_link,in,t} &= v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} &= v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
v^{flow}_{expensive\_supply,t} &= p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
\sum_{t \in t_{cheap}} v^{flow}_{cheap\_supply,t} \cdot 1h &= p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t_{cheap}} \cdot \Delta_{t_{cheap}}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq{} & v^{flow}_{cheap\_link,in,t} \\
0 \leq{} & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq{} & v^{flow}_{expensive\_link,in,t} \\
0 \leq{} & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq{} & v^{flow}_{triangle,in,t} \\
0 \leq{} & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq{} & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq{} & v^{flow}_{expensive\_source,t} \\
0 \leq{} & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq{} & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

2.4 stochastic

The goal of this setup is to test scenarios in the formulation. To that end the setup starts from resolution = fixed and adds stochastics for the demand and the commodity prices. The demand can be high and low and the commodity prices can be expensive or cheap. Combining these scenarios, results in 4 equally likely scenarios for the entire model.

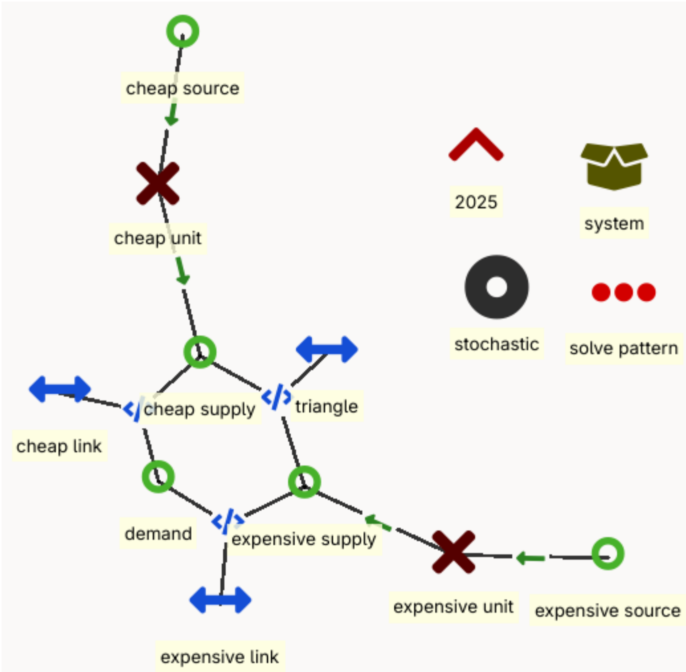

Figure 2.5: stochastic = flexible

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_s p_s^{weight} \cdot \sum_t p_{cheap\_source,t,s}^{other\_operational\_costs} \cdot v_{cheap\_source,t,s}^{flow} + p_{expensive\_source,t,s}^{other\_operational\_costs} \cdot v_{expensive\_source,t,s}^{flow}$$

subject to:

$$\begin{aligned}
p_{demand,t,s}^{flow_profile} =& v_{cheap_link,out,t,s}^{flow} + v_{expensive_link,out,t,s}^{flow} \\
v_{cheap_link,out,t,s}^{flow} =& p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t,s}^{flow} \\
v_{expensive_link,out,t,s}^{flow} =& p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t,s}^{flow} \\
v_{triangle,out,t,s}^{flow} =& p_{triangle}^{efficiency} \cdot v_{triangle,in,t,s}^{flow} \\
v_{cheap_link,in,t,s}^{flow} =& v_{cheap_supply,t,s}^{flow} + v_{triangle,out,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& v_{expensive_link,in,t,s}^{flow} + v_{triangle,in,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t,s}^{flow} \\
v_{cheap_supply,t,s}^{flow} =& p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t,s}^{flow}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \le\, & v^{flow}_{cheap\_link,in,t,s} \\
0 \le\, & v^{flow}_{cheap\_link,out,t,s} && \le p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \le\, & v^{flow}_{expensive\_link,in,t,s} \\
0 \le\, & v^{flow}_{expensive\_link,out,t,s} && \le p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \le\, & v^{flow}_{triangle,in,t,s} \\
0 \le\, & v^{flow}_{triangle,out,t,s} && \le p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \le\, & v^{flow}_{expensive\_supply,t,s} && \le p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \le\, & v^{flow}_{expensive\_source,t,s} \\
0 \le\, & v^{flow}_{cheap\_supply,t,s} && \le p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \le\, & v^{flow}_{cheap\_source,t,s}
\end{aligned}
$$

2.5 reserves

The goal of this setup is to test the reserve requirements. To that end the setup starts from resolution = fixed and adds a reserve requirement to the demand. The lines need to be able to process the reserves and the supply units need to be able to provide reserves. For the proof of concept, we limit ourselves to upward reserves.

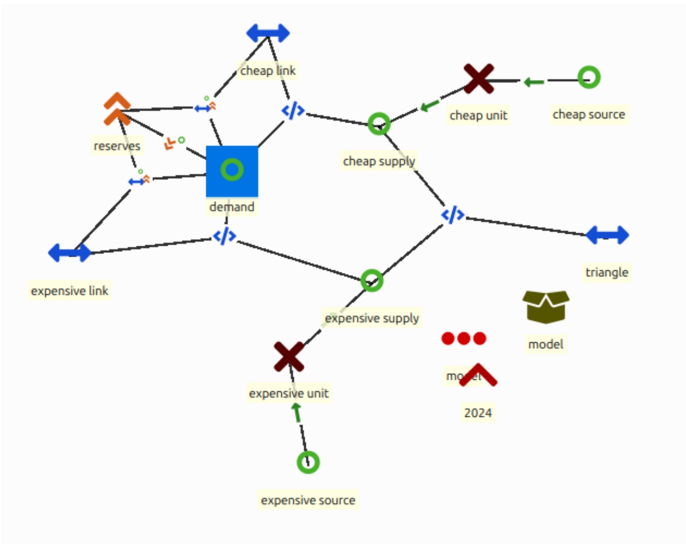

Figure 2.6: reserves

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t} + p^{penalty}_{demand} \cdot v^{slack}_{demand,t}$$

subject to:

$$p^{reserve}_{demand,t} =< v^{flow,reserve}_{cheap\_link,out,t} + v^{flow,reserve}_{expensive\_link,out,t} + v^{slack}_{demand,t}$$

$$p^{flow\_profile}_{demand,t} = v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}$$

$$\begin{aligned}
v^{flow}_{cheap_link,out,t} &= p^{efficiency}_{cheap_link} \cdot v^{flow}_{cheap_link,in,t} \\
v^{flow}_{expensive_link,out,t} &= p^{efficiency}_{expensive_link} \cdot v^{flow}_{expensive_link,in,t} \\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow,reserve}_{cheap_link,out,t} &= p^{efficiency}_{cheap_link} \cdot v^{flow,reserve}_{cheap_link,in,t} \\
v^{flow,reserve}_{expensive_link,out,t} &= p^{efficiency}_{expensive_link} \cdot v^{flow,reserve}_{expensive_link,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{cheap_link,out,t} + v^{flow,reserve}_{cheap_link,out,t} &\leq p^{capacity}_{cheap_link} \cdot p^{existing_links}_{cheap_link} \\
v^{flow}_{expensive_link,out,t} + v^{flow,reserve}_{expensive_link,out,t} &\leq p^{capacity}_{expensive_link} \cdot p^{existing_links}_{expensive_link}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{cheap_link,in,t} &= v^{flow}_{cheap_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive_supply,t} &= v^{flow}_{expensive_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow,reserve}_{cheap_supply,t} &= v^{flow,reserve}_{cheap_link,in,t} \\
v^{flow,reserve}_{expensive_supply,t} &= v^{flow,reserve}_{expensive_link,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{expensive_supply,t} &= p^{efficiency}_{expensive_unit} \cdot v^{flow}_{expensive_source,t} \\
v^{flow}_{cheap_supply,t} &= p^{efficiency}_{cheap_unit} \cdot v^{flow}_{cheap_source,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{expensive_supply,t} + v^{flow,reserve}_{expensive_supply,t} &\leq p^{capacity}_{expensive_unit} \cdot p^{existing_units}_{expensive_unit} \\
v^{flow}_{cheap_supply,t} + v^{flow,reserve}_{expensive_supply,t} &\leq p^{capacity}_{cheap_unit} \cdot p^{existing_units}_{cheap_unit}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq & v^{slack}_{demand,t} \\
0 \leq & v^{flow,reserve}_{cheap\_link,out,t} && \leq p^{max\_reserve}_{cheap\_link} \cdot p^{existing\_links}_{cheap\_link} \\
0 \leq & v^{flow,reserve}_{expensive\_link,out,t} && \leq p^{max\_reserve}_{expensive\_link} \cdot p^{existing\_links}_{expensive\_link} \\
0 \leq & v^{flow,reserve}_{cheap\_supply,t} && \leq p^{max\_reserve}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow,reserve}_{expensive\_supply,t} && \leq p^{max\_reserve}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_links}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_links}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
\end{aligned}
$$

2.6 unit commitment

The goal of this setup is to test the unit commitment functionality. To that end, the setup starts from resolution = fixed but adds on/off state variables to the supply units. The commitment is constrained by the minimum up and down time and there is a cost associated to starting units. Part load efficiency is implemented as well. The startup cost and minimum up and down time are large for the cheap unit to incentivise the model to choose the expensive unit.

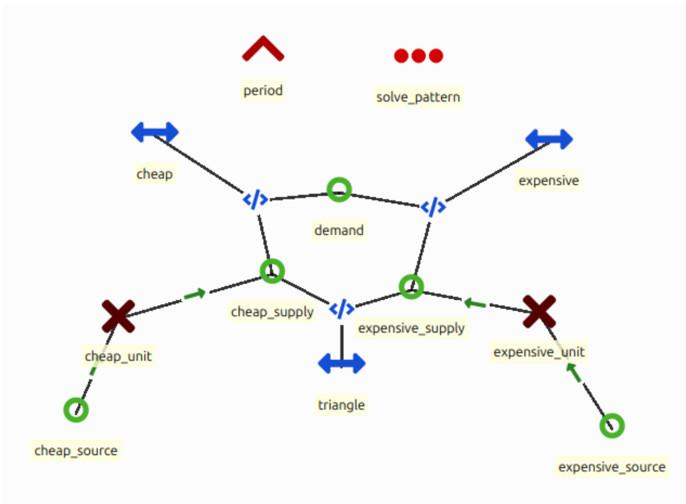

Figure 2.7: Implementation of unit commitment in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = \sum_t p^{startup\_cost}_{cheap\_unit} \cdot v^{on\_up.}_{cheap\_unit,t} + p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} \\
+ p^{startup\_cost}_{expensive\_unit} \cdot v^{on\_up.}_{expensive\_unit,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
\end{aligned}
$$

subject to:

$$
p^{flow\_profile}_{demand,t} = v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} &= p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} &= p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} &= v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} &= v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &= p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
&+ \left(1 - \frac{p^{efficiency}_{expensive\_unit}}{p^{efficiency\_min}_{expensive\_unit}}\right) \cdot p^{capacity\_min}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \cdot v^{on}_{expensive\_unit,t} \\
v^{flow}_{cheap\_supply,t} &= p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t} \\
&+ \left(1 - \frac{p^{efficiency}_{cheap\_unit}}{p^{efficiency\_min}_{cheap\_unit}}\right) \cdot p^{capacity\_min}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \cdot v^{on}_{cheap\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on}_{cheap\_unit,t} &= v^{on}_{cheap\_unit,t-1} + v^{on\_up}_{cheap\_unit,t} - v^{on\_down}_{cheap\_unit,t} \\
v^{on}_{expensive\_unit,t} &= v^{on}_{expensive\_unit,t-1} + v^{on\_up}_{expensive\_unit,t} - v^{on\_down}_{expensive\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on\_up}_{cheap\_unit,t} &\leq 1 - v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{cheap\_unit,t-t'} \\
v^{on\_down}_{cheap\_unit,t} &\leq v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{cheap\_unit,t-t'} \\
v^{on\_up}_{expensive\_unit,t} &\leq 1 - v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{expensive\_unit,t-t'} \\
v^{on\_down}_{expensive\_unit,t} &\leq v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{expensive\_unit,t-t'}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
& v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \cdot v^{on}_{expensive\_unit,t} \\
& v^{flow}_{expensive\_supply,t} && \geq p^{capacity\_min}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \cdot v^{on}_{expensive\_unit,t} \\
0 \leq & v^{flow}_{expensive\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
& v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \cdot v^{on}_{cheap\_unit,t} \\
& v^{flow}_{cheap\_supply,t} && \geq p^{capacity\_min}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \cdot v^{on}_{cheap\_unit,t} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{on}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on\_up}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on\_down}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on}_{expensive\_unit,t} && \leq 1 \\
0 \leq & v^{on\_up}_{expensive\_unit,t} && \leq 1 \\
0 \leq & v^{on\_down}_{expensive\_unit,t} && \leq 1
\end{aligned}
$$

2.7 investments = fixed

The goal of this setup is to test basic investments. To that end the resolution = fixed setup is expanded with investment variables. The expensive unit is already invested in to incentivise the model to keep using the expensive unit. In the formulation below, for brevity, the investment cost is to be interpreted as the cost per unit, as opposed to the cost per MW(h).

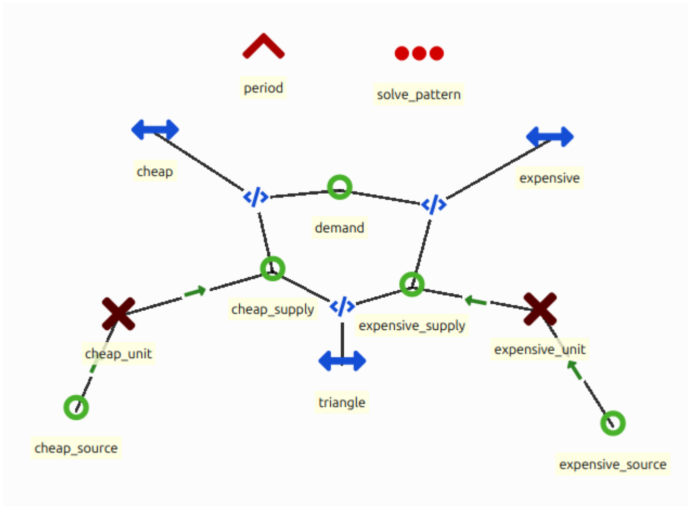

Figure 2.8: Implementation of investment = fixed in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & p^{investment\_cost}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit} + p^{investment\_cost}_{expensive\_unit} \cdot (v^{investment}_{expensive\_unit} - p^{existing\_units}_{expensive\_unit}) \\
& + \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
\\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
\\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
\\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t} \\
\\
v^{flow}_{expensive\_supply,t} \leq& p^{capacity}_{expensive\_unit} \cdot v^{investment}_{expensive\_unit} \\
v^{flow}_{cheap\_supply,t} \leq& p^{capacity}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t} \\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq& v^{flow}_{expensive\_link,in,t} \\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq& v^{flow}_{triangle,in,t} \\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq& v^{flow}_{expensive\_supply,t} \\
0 \leq& v^{flow}_{expensive\_source,t} \\
0 \leq& v^{flow}_{cheap\_supply,t} \\
0 \leq& v^{flow}_{cheap\_source,t} \\
0 \leq& v^{investment}_{cheap\_unit} \\
p^{existing\_units}_{expensive\_unit} \leq& v^{investment}_{expensive\_unit}
\end{aligned}
$$

2.8 storage

The goal of this setup is to test a simple storage formulation. To that end, storage and a renewable unit are added to the resolution = fixed setup. The storage unit is located at the cheap supply side and the renewable is located at the expensive supply side.

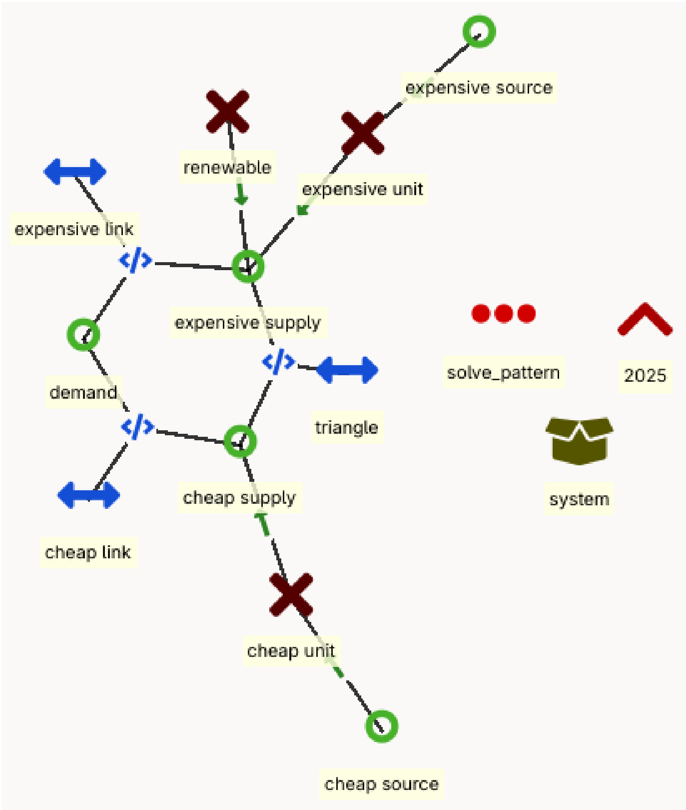

Figure 2.9: Implementation of storage in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
\\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
\\
v^{flow}_{cheap\_link,in,t} + v^{charge}_{storage,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} + p^{efficiency,discharge}_{storage} \cdot v^{discharge}_{storage,t} \\
v^{flow}_{expensive\_supply,t} + v^{flow}_{renewable,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
\\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t} \\
\\
v^{state}_{storage,t} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,t} - v^{discharge}_{storage,t} \\
v^{state}_{storage,1} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,T} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,1} - v^{discharge}_{storage,1}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
\\
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
\\
0 \leq & v^{flow}_{renewable,t} && \leq p^{flow\_profile}_{renewable,t} \cdot p^{capacity}_{renewable} \cdot p^{existing\_units}_{renewable} \\
\\
0 \leq & v^{charge}_{storage,t} && \leq p^{capacity\_charge}_{storage} \cdot p^{existing\_units}_{storage} \\
0 \leq & v^{discharge}_{storage,t} && \leq p^{capacity\_discharge}_{storage} \cdot p^{existing\_units}_{storage} \\
0 \leq & v^{state}_{storage,t} && \leq p^{capacity}_{storage} \cdot p^{existing\_units}_{storage}
\end{aligned}
$$

2.9 rolling horizon

The goal of the setup is to test a rolling horizon. To that end the investment = fixed setup is expanded with a second horizon. Each horizon uses the same formulation. In the first horizon, there is already a small investment in expensive units and the model needs to invest in cheaper units. The second window has a higher demand requiring additional investments but there is also a shift in commodity prices such that the more 'expensive' unit becomes more attractive, yet there are already the investments from the first window in the 'cheaper' units.

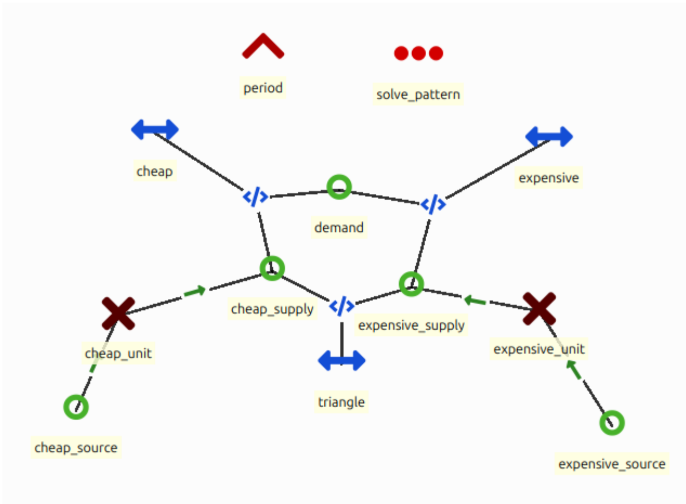

Figure 2.10: Implementation of unit commitment in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & p^{investment\_cost}_{cheap\_unit} \cdot (v^{investment}_{cheap\_unit} - p^{existing\_units}_{cheap\_unit}) + p^{investment\_cost}_{expensive\_unit} \cdot (v^{investment}_{expensive\_unit} - p^{existing\_units}_{expensive\_unit}) \\
& + \sum_{t} p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} &= v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} &= p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} &= p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} &= v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} &= v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &= p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} &= p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &\le p^{capacity}_{expensive\_unit} \cdot v^{investment}_{expensive\_unit}\\
v^{flow}_{cheap\_supply,t} &\le p^{capacity}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 &\le v^{flow}_{cheap\_link,in,t}\\
0 &\le v^{flow}_{cheap\_link,out,t} && \le p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}\\
0 &\le v^{flow}_{expensive\_link,in,t}\\
0 &\le v^{flow}_{expensive\_link,out,t} && \le p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}\\
0 &\le v^{flow}_{triangle,in,t}\\
0 &\le v^{flow}_{triangle,out,t} && \le p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}\\
0 &\le v^{flow}_{expensive\_supply,t}\\
0 &\le v^{flow}_{expensive\_source,t}\\
0 &\le v^{flow}_{cheap\_supply,t}\\
0 &\le v^{flow}_{cheap\_source,t}\\
p^{existing\_units}_{cheap\_unit} &\le v^{investment}_{cheap\_unit}\\
p^{existing\_units}_{expensive\_unit} &\le v^{investment}_{expensive\_unit}
\end{aligned}
$$

2.10 combined system

The goal of this setup is to test systems that combine some of the different features. This setup is a combination of all the previous setups. It is sector coupled for the flexible time steps. The gas sector consists of import, demand

and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. There are stochastics for the renewables and import prices. There is a rolling horizon as well.

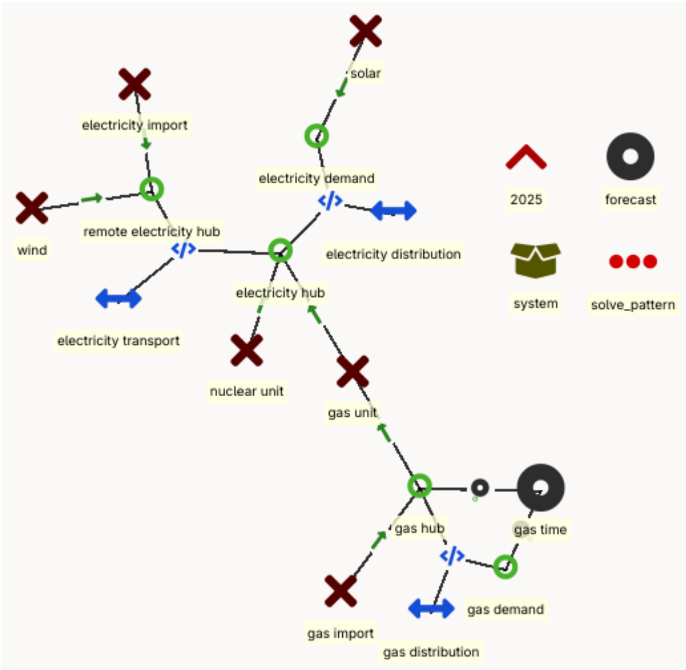

Figure 2.11: combined system

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & \sum_s W_s \sum_{t \in t_{gas}} p^{other\_operational\_costs}_{gas\_import,s,t} \cdot v^{flow}_{gas\_import,s,t} + \sum_t p^{other\_operational\_costs}_{electricity\_import,s,t} \cdot v^{flow}_{electricity\_import,s,t} \\
& + \sum_s W_s \sum_t p^{startup\_cost}_{nuclear\_unit} \cdot v^{on\_up.}_{nuclear\_unit,s,t} + p^{other\_operational\_costs}_{nuclear\_unit} \cdot v^{flow}_{nuclear\_unit,s,t} \\
& + p^{investment\_cost}_{gas\_unit} \cdot (v^{investment}_{gas\_unit} - p^{existing\_units}_{gas\_unit}) + p^{investment\_cost}_{nuclear\_unit} \cdot (v^{investment}_{nuclear\_unit} - p^{existing\_units}_{nuclear\_unit}) \\
& + p^{investment\_cost}_{wind\_unit} \cdot (v^{investment}_{wind\_unit} - p^{existing\_units}_{wind\_unit}) + p^{investment\_cost}_{solar\_unit} \cdot (v^{investment}_{solar\_unit} - p^{existing\_units}_{solar\_unit}) \\
& + p^{investment\_cost}_{battery} \cdot (v^{investment}_{battery} - p^{existing\_units}_{battery})
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{electricity\_demand,t} =& v^{flow}_{electricity\_distribution,out,s,t} + v^{flow}_{solar\_unit,s,t} \\
p^{flow\_profile}_{gas\_demand,t} =& v^{flow}_{gas\_distribution,out,s,t} \\
v^{flow}_{gas\_import,s,t} =& v^{flow}_{gas\_distribution,in,s,t} + v^{flow}_{gas\_unit,in,s,t} \\
v^{flow}_{electricity\_distribution,in,s,t} =& v^{flow}_{electricity\_transport,out,s,t} + v^{flow}_{nuclear\_unit,s,t} + v^{flow}_{gas\_unit,out,s,t} \\
& - v^{charge}_{battery,s,t} + p^{efficiency,discharge}_{battery} \cdot v^{discharge}_{battery,s,t} \\
v^{flow}_{electricity\_transport,in,s,t} =& v^{flow}_{wind\_unit,s,t} + v^{flow}_{electricity\_import,out,s,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{gas\_distribution,out,s,t} =& p^{efficiency}_{gas\_distribution} \cdot v^{flow}_{gas\_distribution,in,s,t} \\
v^{flow}_{electricity\_distribution,out,s,t} =& p^{efficiency}_{electricity\_distribution} \cdot v^{flow}_{electricity\_distribution,in,t} \\
v^{flow}_{electricity\_transport,out,s,t} =& p^{efficiency}_{electricity\_transport} \cdot v^{flow}_{electricity\_transport,in,s,t}
\end{aligned}
$$

$$
\sum_{t \in t_{gas}} v^{flow}_{gas\_unit,out,s,t} \cdot 1h = p^{efficiency}_{gas\_unit} \cdot v^{flow}_{gas\_unit,in,s,t_{gas}} \cdot \Delta_{t_{gas}}
$$

$$
\begin{aligned}
v^{on}_{nuclear\_unit,s,t} =& v^{on}_{nuclear\_unit,s,t-1} + v^{on\_up}_{nuclear\_unit,s,t} - v^{on\_down}_{nuclear\_unit,s,t} \\
v^{on\_up}_{nuclear\_unit,s,t} \leq& 1 - v^{on}_{nuclear\_unit,s,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{nuclear\_unit,s,t-t'} \\
v^{on\_down}_{nuclear\_unit,s,t} \leq& v^{on}_{nuclear\_unit,s,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{nuclear\_unit,s,t-t'}
\end{aligned}
$$

$$
\begin{aligned}
v^{state}_{storage,s,t} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,s,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,s,t} - v^{discharge}_{storage,s,t} \\
v^{state}_{storage,s,1} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,s,T} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,s,1} - v^{discharge}_{storage,s,1}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq v^{flow}_{gas\_distribution,out,s,t} &\leq p^{capacity}_{gas\_distribution} \cdot p^{existing\_units}_{gas\_distribution}\\
0 \leq v^{flow}_{electricity\_distribution,out,s,t} &\leq p^{capacity}_{electricity\_distribution} \cdot p^{existing\_units}_{electricity\_distribution}\\
0 \leq v^{flow}_{electricity\_transport,out,s,t} &\leq p^{capacity}_{electricity\_transport} \cdot p^{existing\_units}_{electricity\_transport}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq v^{flow}_{gas\_import,s,t} &\leq p^{capacity}_{gas\_import} \cdot p^{existing\_units}_{gas\_import}\\
0 \leq v^{flow}_{electricity\_import,s,t} &\leq p^{capacity}_{electricity\_import} \cdot p^{existing\_units}_{electricity\_import}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{nuclear\_unit,s,t} &\leq p^{capacity}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit} \cdot v^{on}_{nuclear\_unit,s,t}\\
v^{flow}_{nuclear\_unit,s,t} &\geq p^{capacity\_min}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit} \cdot v^{on}_{nuclear\_unit,st}\\
0 \leq v^{on}_{nuclear\_unit,s,t} &\leq 1\\
0 \leq v^{on\_up}_{nuclear\_unit,s,t} &\leq 1\\
0 \leq v^{on\_down}_{nuclear\_unit,s,t} &\leq 1
\end{aligned}
$$

$$
\begin{aligned}
0 \leq v^{flow}_{gas\_unit,in,s,t} &\leq \frac{1}{p^{efficiency}_{gas\_unit}} \cdot p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit}\\
0 \leq v^{flow}_{gas\_unit,out,s,t} &\leq p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq v^{flow}_{solar\_unit,s,t} &\leq p^{flow\_profile}_{solar\_unit,s,t} \cdot p^{capacity}_{solar\_unit} \cdot v^{investment}_{solar\_unit}\\
0 \leq v^{flow}_{wind\_unit,s,t} &\leq p^{flow\_profile}_{wind\_unit,s,t} \cdot p^{capacity}_{wind\_unit} \cdot v^{investment}_{wind\_unit}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq v^{charge}_{battery,s,t} &\leq p^{capacity\_charge}_{battery} \cdot v^{investment}_{battery}\\
0 \leq v^{discharge}_{battery,s,t} &\leq p^{capacity\_discharge}_{storage} \cdot v^{investment}_{battery}\\
0 \leq v^{state}_{battery,s,t} &\leq p^{capacity}_{storage} \cdot v^{investment}_{battery}
\end{aligned}
$$

$$
\begin{aligned}
p^{existing\_units}_{gas\_unit} &\leq v^{investment}_{gas\_unit}\\
p^{existing\_units}_{nuclear\_unit} &\leq v^{investment}_{nuclear\_unit}\\
p^{existing\_units}_{solar\_unit} &\leq v^{investment}_{solar\_unit}\\
p^{existing\_units}_{wind\_unit} &\leq v^{investment}_{wind\_unit}\\
p^{existing\_units}_{battery} &\leq v^{investment}_{battery}
\end{aligned}
$$

Appendix A

ines spec

The intended way to inspect the ines spec is through Spine Toolbox. For reference, a static table is provided below.

Appendix B

Formulation ines [work in progress]

From the certification process, it is possible to piece together the overall intended formulation of the ines spec. As the certification process is currently only a proof of concept, this formulation is not yet complete and as such a work in progress.

The ines spec is disentangled here in the typical formulation of a linear program: indices, parameters, variables, objective and constraints. The formulation parameters and variables are formatted as $p^{name}_{indices}$ and $v^{name}_{indices}$ respectively. The names correspond directly to the input parameters in the ines spec, unless otherwise specified in the preprocessing section. The indices are always the most complex possible datastructure making the equations valid for all possible input values.

B.1 Indices

set groups of units, connections, etc.

set_{unit} set that a unit belongs to

$set_{connection}$ set that a connection belongs to

set_{node} set that a node belongs to

$period$ investment periods in the system

$time$ time steps (always continuous within period or rolling horizon, weight is used for representative days)

$time_{set}$ time steps in set

$period_{time}$ period corresponding to time

$unit$ units in the system

$unit\_\_to\_node$ unit to nodes in the system

$unit_{unit\_\_to\_node}$ unit in the unit__to_node entity

$node\_\_to\_unit$ unit to nodes in the system

$unit_{node\_\_to\_unit}$ unit in the unit__to_node entity

B.2 Parameters

The parameters are formatted as $p^{parameter\_name}_{sets}$ with the parameter name from the table below and the set based on the maximum of the valid types.

node__to_unit has the same parameters as unit__to_node so that is not repeated here

$p^{interest\_rate}_{unit}$ discount rate related to a unit

$p^{startup\_cost}_{unit,period,time}$ startup cost

$p^{shutdown\_cost}_{unit,period,time}$ shutdown cost

$p^{online\_cost}_{unit,period,time}$ online cost

$p^{investment\_cost}_{unit\_\_to\_node,period}$ investment cost of a unit flow in a certain period

$p^{fixed\_cost}_{unit\_\_to\_node,period}$ FOM cost of a unit flow

$p^{other\_operational\_cost}_{un,p,t}$ VOM cost of a unit flow

$p^{ramp\_cost}_{unit\_\_to\_node,period}$ the ramp cost is specifically described to be the cost of ramping up (not ramping down)

B.3 Variables

According to the ines-output spec, for a *node__to_unit*, there is a capacity, a flow per period and a flow per time step.

node__to_unit is the same as unit__to_node so it is not repeated here.

$v^{commit}_{un,p,t}$ committed units

$v^{commit\_up}_{un,p,t}$ units that started up

$v^{commit\_down}_{un,p,t}$ units that shut down

$v^{investment}_{unit\_\_to\_node,period}$ additional capacity of installed unit__to_node in a period

$v^{capacity}_{unit\_\_to\_node,period}$ capacity of unit__to_node in each period

$v^{flow}_{unit\_\_to\_node,period,time}$ flow of unit__to_node in each time in each period

$v^{rampup}_{unit\_\_to\_node,period,time}$ ramp up of unit__to_node in each time in each period

B.4 Objective

$$
\begin{aligned}
\min \\
&+ \sum_{u \in unit} \sum_{p \in period} p^{economic\_other}_{u,p} \cdot \sum_{t \in time_p} \Delta_t \cdot p^{online\_cost}_{un,p} \cdot v^{commit}_{un,p,t} \\
&+ \sum_{u \in unit} \sum_{p \in period} p^{economic\_other}_{u,p} \cdot \sum_{t \in time_p \notin 1} p^{startup\_cost}_{un,p} \cdot v^{commit\_up}_{un,p,t} \\
&+ \sum_{u \in unit} \sum_{p \in period} p^{economic\_other}_{u,p} \cdot \sum_{t \in time_p \notin 1} p^{shutdown\_cost}_{un,p} \cdot v^{commit\_down}_{un,p,t} \\
\\
&+ \sum_{un \in unit\_\_to\_node} \sum_{p \in period} p^{economic\_investment}_{unit[un],p} \cdot p^{investment\_cost}_{un,p} \cdot v^{investment}_{un,p} \\
&+ \sum_{un \in unit\_\_to\_node} \sum_{p \in period} p^{economic\_fixed}_{unit[un],p} \cdot p^{fixed\_cost}_{un,p} \cdot v^{capacity}_{un,p} \\
&+ \sum_{un \in unit\_\_to\_node} \sum_{p \in period} p^{economic\_other}_{unit[un],p} \cdot \sum_{t \in time_p} \Delta_t \cdot p^{other\_operational\_cost}_{un,p,t} \cdot v^{flow}_{un,p,t} \\
&+ \sum_{un \in unit\_\_to\_node} \sum_{p \in period} p^{economic\_other}_{unit[un],p} \cdot \sum_{t \in time_p \notin 1} p^{ramp\_cost}_{un,p} \cdot v^{ramp\_up}_{un,p,t} \\
\\
&+ \sum_{nu \in node\_\_to\_unit} \sum_{p \in period} p^{economic\_investment}_{unit[nu],p} \cdot p^{investment\_cost}_{nu,p} \cdot v^{investment}_{nu,p} \\
&+ \sum_{nu \in node\_\_to\_unit} \sum_{p \in period} p^{economic\_fixed}_{unit[nu],p} \cdot p^{fixed\_cost}_{nu,p} \cdot v^{capacity}_{nu,p} \\
&+ \sum_{nu \in node\_\_to\_unit} \sum_{p \in period} p^{economic\_other}_{unit[nu],p} \cdot \sum_{t \in time_p} \Delta_t \cdot p^{other\_operational\_cost}_{nu,p,t} \cdot v^{flow}_{nu,p,t} \\
&+ \sum_{un \in node\_\_to\_unit} \sum_{p \in period} p^{economic\_other}_{unit[nu],p} \cdot \sum_{t \in time_p \notin 1} p^{ramp\_cost}_{nu,p,t} \cdot v^{ramp\_up}_{nu,p,t}
\end{aligned}
$$

B.5 Reserves

$\forall n \in node\_\_reserve:$

$$
\begin{aligned}
&\sum_{u \in n} r^{+}_{u,t} + \sum_{u \in su \cup n} r^{c,+}_{u,t} + r^{d,+}_{u,t} + \sum_{u \in dlu \cup n} r^{l,+}_{u,t} + \sum_{u \in du \cup n} r^{ll}_{u,t} \\
&- \sum_{u \in du \cup n} F^{dres} \cdot P^{peak}_{u,Y_t} - \sum_{u \in ru \cup n} F^{res} \cdot c_{u,Y_t} = 0 \\
&\sum_{u \in n} r^{-}_{u,t} + \sum_{u \in su \cup n} r^{c,-}_{u,t} + r^{d,-}_{u,t} + \sum_{u \in dlu \cup n} r^{l,-}_{u,t} \\
&- \sum_{u \in du \cup n} F^{dres} \cdot P^{peak}_{u,Y_t} - \sum_{u \in ru \cup n} F^{res} \cdot c_{u,Y_t} = 0
\end{aligned}
$$

B.6 Unit Constraints

$$
\begin{aligned}
v_{un,p,t}^{ramp\_up} &> 0 \\
v_{un,p,t}^{ramp\_up} &> v_{un,p,t}^{flow} - v_{un,p,t-1}^{flow} \\
v_{un,p,t}^{ramp\_down} &> 0 \\
v_{un,p,t}^{ramp\_down} &> v_{un,p,t-1}^{flow} - v_{un,p,t}^{flow}
\end{aligned}
$$

Appendix C

Extended Certification process

The certification setups from Chapter 2 are selected for a proof of concept. The full certification process consists of:

equation tests are a systematic group of tests to test each equation of energy system models individually, starting from a simple energy balance on one node with 1 demand unit and on supply unit

method tests are tests that cover multiple equations to show off a certain feature (various examples can be found in Chapter 2)

system tests are more comprehensible, similar to IEEE networks (an example can be found in the final setup of Chapter 2)

For the equation tests, inspiration can be taken from the different formulation in the proof of concept of the certification process. For the method tests, table C.1 is an indicative expansion of the certification process. The system tests are a point of discussion for the consortium of the ines spec.

Table C.1: Expansion of the certification process

name	description
network = copper plate	Energy balance where the demand equals the supply for a single time step. Two units where one is cheaper than the other.
network = radial	Two supply units, each connected with a directional line to the demand. The cheaper unit has the direction from the demand to the supply so the setup has no choice but to choose the more expensive unit.

network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular network with the two supply units and demand unit. The cheapest unit should be chosen.
resolution = fixed	Same setup but with a time series for all units with the same resolution.
resolution = variable	Same setup but the time series has a variable resolution.
resolution = flexible	Same setup but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
stochastic	Meshed network, fixed resolution, expensive and cheap unit. The units have the same stochastics.
stochastic = flexible	Same setup but the units have different stochastics.
CO2 cap	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit produces CO2 and is limited by the cap.
reserves	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
cogeneration	Meshed network, fixed resolution, two supply units and two demand units but one one supply is cogeneration and one demand is for heat. The setup is heat driven as only the electricity grid has two supply units.
ramping	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit is not able to ramp. The demand profile is either an elevated sine or piecewise linear _/.
flow = fixed	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce a certain flow.

flow = limited	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce within certain limits. This test also represents curtailment.
storage = intra	Meshed network, fixed resolution, one storage unit and one supply unit. The supply unit has a fixed output that does not correspond to the demand. The storage has to be used to be feasible.
storage = inter	Same setup but there are multiple periods.
part load efficiency	Meshed network, fixed resolution, expensive and cheap unit. The partload efficiency of the cheap unit makes it that the overall cost is higher than the expensive unit.
MUT	Meshed network, fixed resolution, expensive storage and cheap unit. By setting the minimum production equal to the maximal production the output of the unit becomes fixed. By setting the minimum up time (MUT) equal to the lifetime of the unit, whenever the unit delivers an output, that output will need to be maintained for the remainder of the simulation. That means that there will appear horizontal lines for each level of required output. To make sure that the system works, a storage unit is used to get rid of the excess power. The storage unit is very expensive such that it would be avoided if it would not be needed.
MDT	Same setup, but there is a moment where the demand is zero. Normally, the plant would shut down, but due to the MDT it needs to remain on.
economic representation	Meshed network, fixed resolution, expensive and cheap unit. The numbers will be different from other setups due to the economic representation.

investment = fixed	Meshed network, fixed resolution, expensive and cheap unit. Expensive unit is already invested in.
investment = cumulative	Same setup but a cumulative cap
investment = multi year	Same setup but multiple periods.
retirement = fixed	Same setup Expensive unit is already invested in but is sunset in the second period.
retirement = economic	Same setup but the retirement is not fixed.
representative periods	
blended representative periods	
rolling horizon	Same setup but the periods are separated.
benders decomposition	Same setup but the investment and operation are separated from each other.
mini Europe	A miniature version of europe with very limited temporal aspects (a day = 6 hours, a season is 1 day). Each country has a demand but only its most prominent technologies. Complex setup with storage, cogeneration, CO2 cap, ramping, flow limits, part load efficiency, MUT/MDT.